

Eigenstructure of **A** ✱ work in appended PDF. (using maple)

eigenvalues: $\lambda_1 = u - a$, $\lambda_2 = u + a$, $\lambda_3 = \lambda_4 = u$

right eigenvectors: $\vec{r}_1 = \begin{bmatrix} -p/a \\ 1 \\ 0 \\ -ap \end{bmatrix}$, $\vec{r}_2 = \begin{bmatrix} 1 \\ a/p \\ 0 \\ a^2 \end{bmatrix}$, $\vec{r}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{r}_4 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

left eigenvectors: $\vec{l}_1 = \left[0, \frac{1}{2}, 0, -\frac{1}{2ap} \right]$
 $\vec{l}_2 = \left[0, \frac{p}{2a}, 0, \frac{1}{2a^2} \right]$
 $\vec{l}_3 = \left[1, 0, 0, -\frac{1}{a^2} \right]$
 $\vec{l}_4 = \left[0, 1, 0, 0 \right]$

Eigenstructure of **B**

eigenvalues: $\lambda_1 = v - a$, $\lambda_2 = v + a$, $\lambda_3 = \lambda_4 = v$

right eigenvectors: $\vec{r}_1 = \begin{bmatrix} -p/a \\ 0 \\ 1 \\ -ap \end{bmatrix}$, $\vec{r}_2 = \begin{bmatrix} 1 \\ 0 \\ a/p \\ a^2 \end{bmatrix}$, $\vec{r}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $\vec{r}_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

left eigenvectors: $\vec{l}_1 = \left[0, 0, \frac{1}{2}, -\frac{1}{2ap} \right]$
 $\vec{l}_2 = \left[0, 0, \frac{p}{2a}, \frac{1}{2a^2} \right]$
 $\vec{l}_3 = \left[1, 0, 0, -\frac{1}{a^2} \right]$
 $\vec{l}_4 = \left[0, 1, 0, 0 \right]$

Eigenstructure of

A

eigenvalues: $\lambda_1 = u-a$, $\lambda_2 = u+a$, $\lambda_3 = \lambda_4 = u$ (Same as **A**)

right eigenvectors:

$$\vec{r}_1 = \left[1, u-a, v, \frac{(-2au + u^2 + v^2)\gamma - u^2 + 2au - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_2 = \left[1, u+a, v, \frac{(2au + u^2 + v^2)\gamma - u^2 - 2au - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_3 = \left[1, u, 0, \frac{u^2}{2} - \frac{v^2}{2} \right]^T$$

$$\vec{r}_4 = \left[0, 0, 1, v \right]^T$$

left eigenvectors:

$$\vec{l}_1 = \left[\frac{u^2(\gamma-1) + 2au + v^2(\gamma-1)}{4a^2}, \frac{-\gamma u - a + u}{2a^2}, \frac{v(\gamma-1)}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_2 = \left[\frac{u^2(\gamma-1) - 2au + v^2(\gamma-1)}{4a^2}, \frac{-\gamma u + a + u}{2a^2}, \frac{v(\gamma-1)}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_3 = \left[\frac{2a^2 - (u^2 + v^2)(\gamma-1)}{2a^2}, \frac{u(\gamma-1)}{a^2}, \frac{v(\gamma-1)}{a^2}, \frac{-\gamma+1}{a^2} \right]$$

$$\vec{l}_4 = \left[-\frac{(u^2 + v^2)v(\gamma-1)}{2a^2}, \frac{uv(\gamma-1)}{a^2}, \frac{a^2 + v^2(\gamma-1)}{a^2}, -\frac{v(\gamma-1)}{a^2} \right]$$

Eigenstructure of **B**

eigenvalues: $\lambda_1 = v-a$, $\lambda_2 = v+a$, $\lambda_3 = \lambda_4 = v$ (Same as **B**)

right eigenvectors:

$$\vec{r}_1 = \left[1, u, v-a, \frac{(-2av + u^2 + v^2)\gamma - u^2 + 2av - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_2 = \left[1, u, v+a, \frac{(2av + u^2 + v^2)\gamma - u^2 - 2av - v^2 + 2a^2}{2\gamma - 2} \right]^T$$

$$\vec{r}_3 = \left[1, 0, v, -\frac{u^2}{2} + \frac{v^2}{2} \right]^T$$

$$\vec{r}_4 = \left[0, 1, 0, u \right]^T$$

left eigenvectors:

$$\vec{l}_1 = \left[\frac{v^2(\gamma-1) + 2av + u^2(\gamma-1)}{4a^2}, \frac{u(\gamma-1)}{2a^2}, \frac{-\gamma v - a + v}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_2 = \left[\frac{v^2(\gamma-1) - 2av + u^2(\gamma-1)}{4a^2}, \frac{u(\gamma-1)}{2a^2}, \frac{-\gamma v + a + v}{2a^2}, \frac{\gamma-1}{2a^2} \right]$$

$$\vec{l}_3 = \left[\frac{2a^2 - (u^2 + v^2)(\gamma-1)}{2a^2}, \frac{u(\gamma-1)}{a^2}, \frac{v(\gamma-1)}{a^2}, \frac{-\gamma+1}{a^2} \right]$$

$$\vec{l}_4 = \left[-\frac{(u^2 + v^2)u(\gamma-1)}{2a^2}, \frac{a^2 + u^2(\gamma-1)}{a^2}, \frac{uv(\gamma-1)}{a^2}, -\frac{u(\gamma-1)}{a^2} \right]$$